iRpgFormatter for RDi

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Overview

The iRpgFormatter formats the declarations, calculations, comments and embedded SQL statements of RPGLE or SQLRPGLE sources:

- Execute from Remote System filters.
- Execute from Project Explorer items.
- Execute from Lpex view.
- Execute on save.
- Optionally formats the calculations with the IBM formatter (only in Lpex editor).

iRpgFormatter is an open source plug-in for RDi 9.6.0.13+. It formats free-form RPG IV source code to improve readability and consistency.

Feature list:

- Formats free-form RPG IV source code.
- Indents the calculations by the nesting of their blocks and the comment lines like the following statement.
- Formats embedded SQL statements according to the RDi preferences page *Free-form SQL Formatting*.
- Formats the branches of conditional compiling (`/IF`, `/ELSE`, `/ENDIF`) like the compiler.
- Formats selected lines like the whole source.
- Various formatting options:
  - Casing style.
  - Add/remove const() keyword.
  - Break name on case change.
  - Sort const/value to end.
  - Replace dcl-pi name with *N.
  - Remove end-proc name.
  - Unindent compiler directives.

  - Outer parentheses of conditions (experimental).
  - Casing of operation codes and built-in functions (experimental).
  - Casing of special words of the calculations (experimental).
- Optionally formats the calculations with the IBM formatter.
- Optionally format source on save.

iRpgFormatter is an Open Source project that is hosted on github.com.

Limitations

The formatter does not have an Abstract Syntax Tree (AST), which usually is required for formatting source code. But it has basic knowledge about functions, keywords, special words, data types and the blocks of the calculations.

Most of the time it produces the expected result. But there are edge cases where the result is unexpected. Unexpected results can occur if the structure of the source code is not completely visible in the source member, but hidden in includes or manipulated by conditional compiling.

For example:

```
/if defined(DEFINE_PROCPTR)
dcl-pr SrvPgmLayout_new ind extproc(g_pLayout_new);
/else
dcl-pr SrvPgmLayout_new ind extproc('SrvPgmLayout_new');
/endif
  i_pSuper   like(LOG4RPG_pLayout_t) const;
  io_pHandle pointer;
  i_propList like(LOG4RPG_pPropertyList_t) const;
  i_prefix   like(LOG4RPG_prefix_t) const;
end-pr;
```

The problem of the example above is that there are two `dcl-pr` statements in a row. The formatter handles them like the compiler: each branch of an `/IF` group starts with the blocks that are open at `/IF`. If the branches end with different blocks, the formatter shows a warning, because the indentation of the following lines may be wrong.

Menu Options

The *iFormat* menu option is in the *Source* sub-menu. You can map a user key action to it on the **LPEX Editor** – **User Key Actions** page. If lines are selected, only these lines are formatted, indented like in the whole source.

Preferences

The preferences page has been placed next to the IBM *Formatter* preferences page and is labeled as *iRpgFormatter*. The page *iRpgFormatter* at the root of the preferences leads there.

Play with the settings and watch the preview panel to understand the formatting options.

Format declarations (formerly *Execute iRPG formatter*) formats the control options, the declarations and the compiler directives. *Format calculations* (formerly *Execute IBM formatter*) selects, who formats the calculations, the comment lines in front of them and the embedded SQL statements: *Off*, *IBM formatter* or *iRPGFormatter*.

With *IBM formatter*, the preview shows the calculations as iRPGFormatter would format them, because the IBM formatter cannot run in the preview.

Embedded SQL statements are formatted according to the RDi preferences page *Free-form SQL Formatting*, which is linked in the group *IBM Settings*. In RDi 9.6 they are only indented.

The following tabs contain lists of entries used for case-sensitive formatting:

- Data types
- Declaration types
- Keywords
- Special Words
- Operation Codes
- Built-in Functions

The built-in lists correspond to the ILE RPG Reference for IBM i 7.6 as of 2026-10-28.

Profiles exported with a former version can still be imported. Settings added later keep their current values.

A more detailed description of the settings can be found on the iRpgFormatter project page on GitHub.

Value "IBM formatter" of option "Format calculations" only works if the formatter is started in the Lpex editor.

Change Log 1.0.7.r (07.06.2026)

Plug-in: 9.6.0.13+

- Fix `Parents stack contains unexpected parent: dcl-ds...` on statements containing the `likerec` keyword.
- Fix incorrectly committed source code.

Change Log 1.0.6.r (07.06.2026)

Plug-in: 9.6.0.13+

- Fix `Parents stack contains unexpected parent: dcl-ds...` on statements containing the `likerec` keyword.

Change Log 1.0.5.r (01.06.2026)

Plug-in: 9.6.0.13+

- Fix `UnexpectedTokenException` when formatting keywords.

Change Log 1.0.4.r (25.05.2026)

Plug-in: 9.6.0.13+

- Fix `ClassNotFoundException` in RDi 9.6.
- Applied experimental edge case fix: compiler conditions.
- Fix formatting compiler directives.
- Allow spaces between function name and parameter list.
- Fix setting maximum line length.

Change Log 1.0.3.r (12.05.2026)

Plug-in: 9.6.0.13+

- Fix `Parents stack contains unexpected parent: dcl-pi *n end-pi;` parser error.
- Fix `Line n not found.` error in Lpex editor when positioning the cursor at the end of a formatting action.

Change Log 1.0.2.r (12.05.2026)

Plug-in: 9.6.0.13+

- Fix `Unexpected token of type: FUNCTION` exception.

Change Log 1.0.1.r (12.05.2026)

Plug-in: 9.6.0.13+

- Add formatting of compile time arrays.

Change Log 1.0.0.r (12.05.2026)

Plug-in: 9.6.0.13+

- Initial version

iRpgFormatter On The Internet

Project page on GitHub	https://github.com/tools-400/irpgformatter-plugin/
Homepage	https://tools-400.github.io/irpgformatter/
RDi 9.6+ Update Site	https://tools-400.github.io/irpgformatter/update-site/eclipse/rdi8.0/
Eclipse Marketplace	https://marketplace.eclipse.org/content/irpgformatter-rdi-96013/
RDi 9.6 Beta Update Site	https://tools-400.github.io/irpgformatter/beta-version/update-site/eclipse/rdi8.0/